

HUMBER ENGINEERING

MENG-3020

SYSTEMS MODELING & SIMULATION

LECTURE 7

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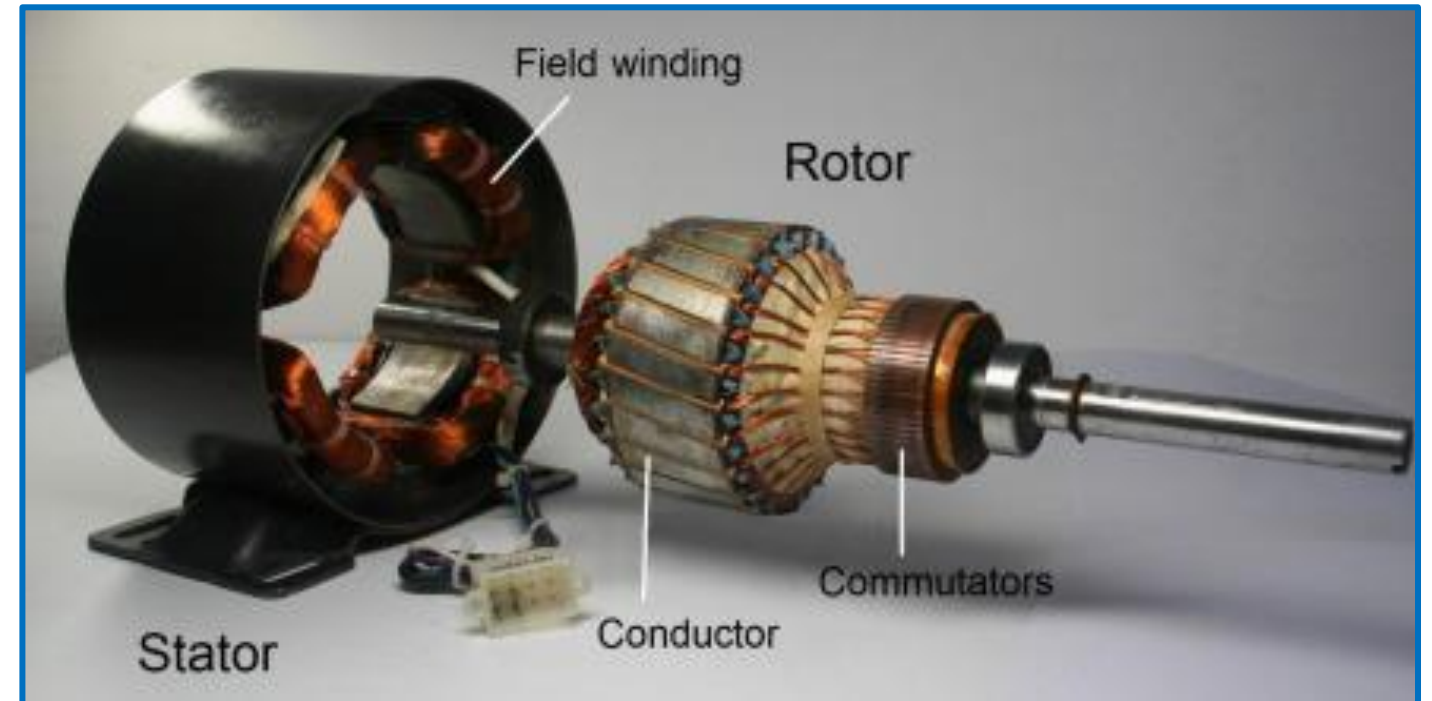
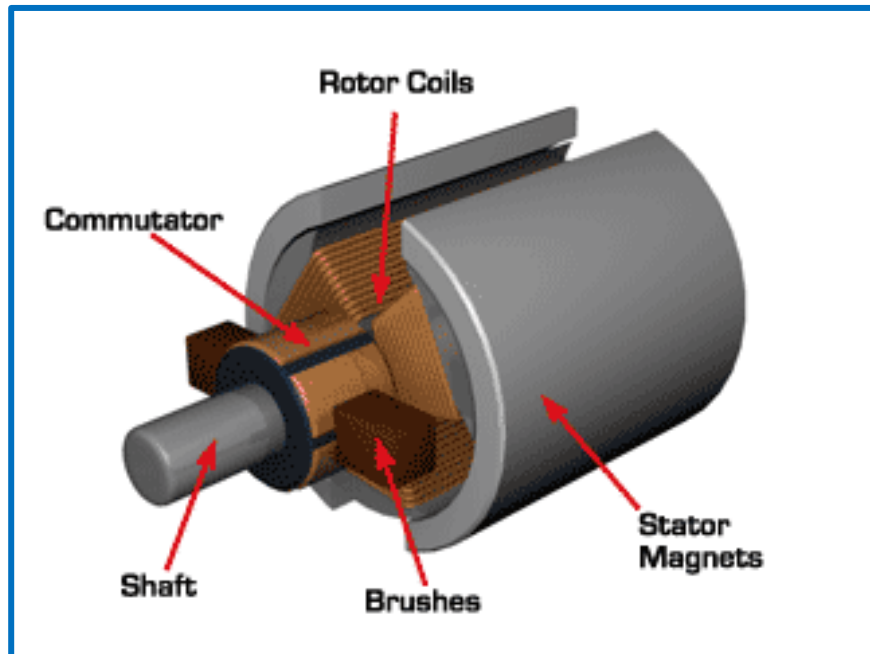
Modeling of Electromechanical Systems

- Basic Characteristics of DC Motors
 - Armature-Controlled DC Motor
 - Field-Controlled DC Motor
 - Differential Equation Model
 - Block Diagram Model
 - Transfer Function Model
 - State-space Model

Modeling of Electromechanical Systems

□ Basic DC Motor

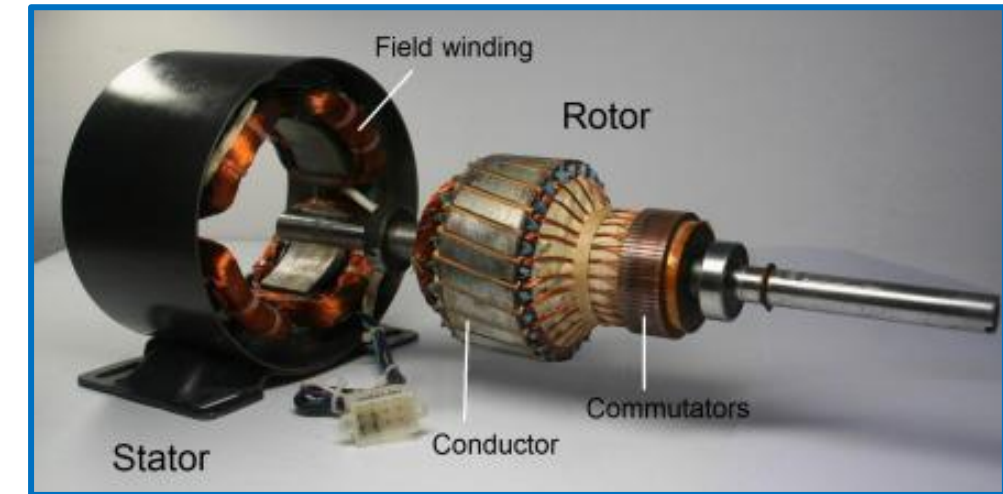
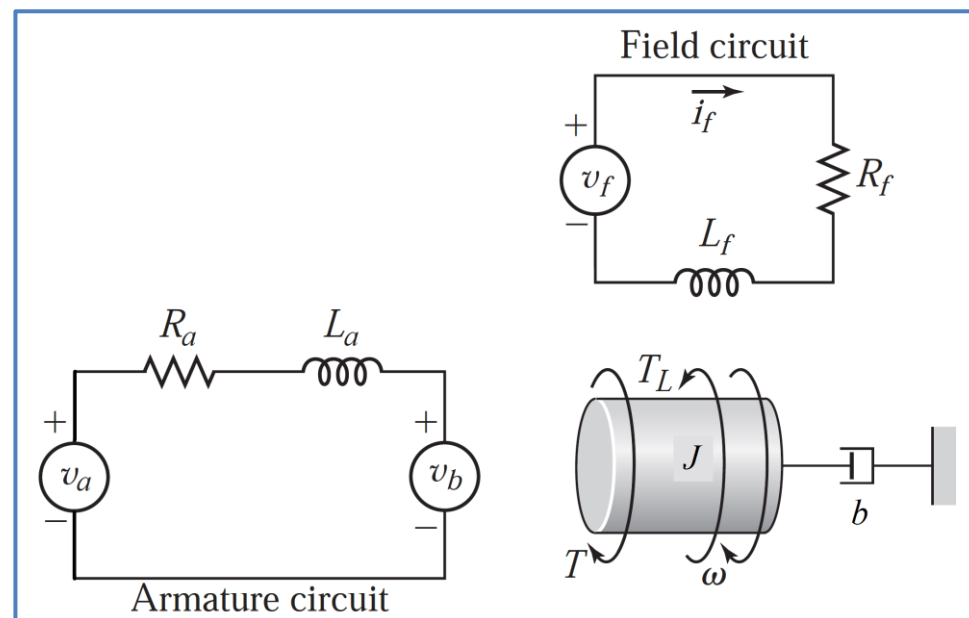
- **DC motor** is basically an **electro-mechanical conversion** device that converts **electric energy** into **mechanical energy**.
- **DC motors** have five principal components:
 - Stator
 - Field System
 - Armature or Rotor
 - Commutator
 - Brushes



Modeling of Electromechanical Systems

□ Model of DC Motors

- **DC motors** can be modeled by the following equivalent electro-mechanical model by considering two subsystems:
- **Electrical Subsystem**
 - Armature circuit, including the armature winding resistance and inductance
 - Field circuit, including the field winding resistance and inductance
- **Mechanical Subsystem**
 - Inertia due to the load as well as the armature inertia.
 - Damping can be present because of shaft bearings or load damping, such as with a fan or pump.



R_a = Armature Resistance, Ω

L_a = Armature Inductance, H

R_f = Field Resistance, Ω

L_f = Field Inductance, H

v_a = Applied Armature Voltage, V

v_b = Back-emf, V

ω = Angular velocity, rad/sec

τ = Torque developed by the motor, $N.m$

J = Moment of inertia of the motor and load referred to the motor shaft,

b = Viscous friction coefficient of the motor and load referred to the motor shaft,

Modeling of Electromechanical Systems

□ Control of DC Motor

- Two general methods to control speed of DC motor:

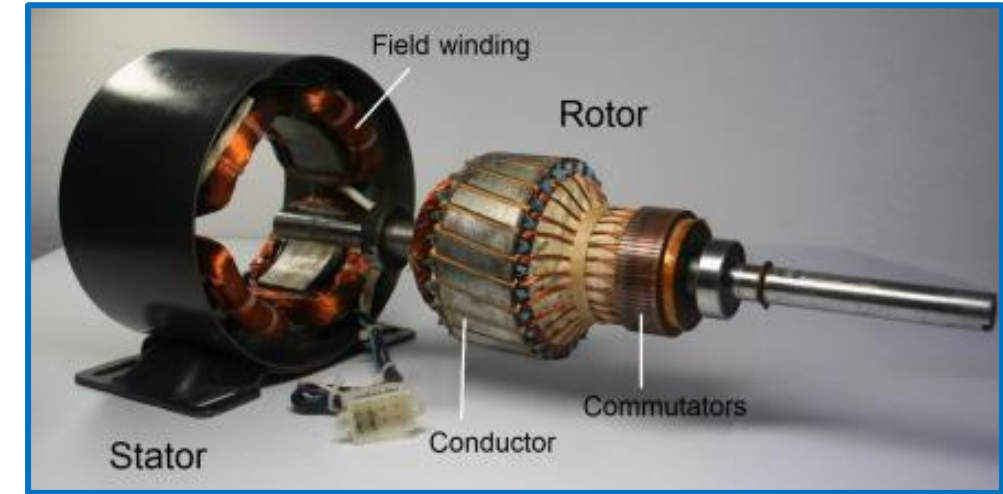


- Armature-Controlled DC Motor**

- Field current i_f and field flux ϕ_f is constant.
- Using permanent magnet

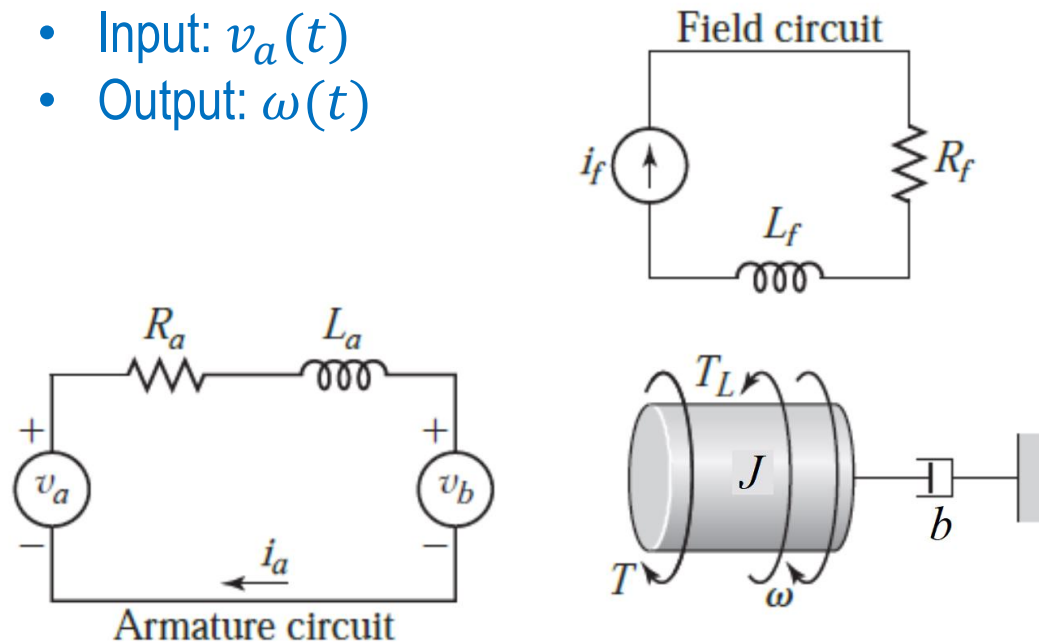
- Field-Controlled DC Motor**

- Armature current i_a is constant



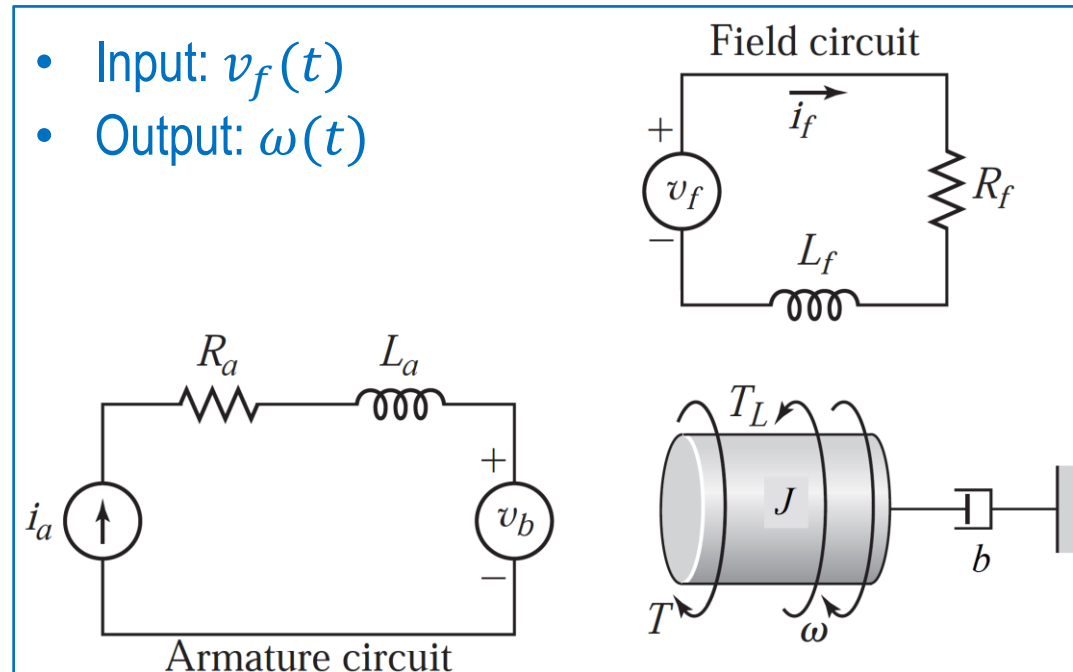
Armature-Controlled DC Motor

- Input: $v_a(t)$
- Output: $\omega(t)$



Field-Controlled DC Motor

- Input: $v_f(t)$
- Output: $\omega(t)$



Modeling of Armature-Controlled DC Motor

- The **torque** τ developed by the motor is proportional to the product of the armature current i_a and the magnetic flux ϕ_f .

$$\tau \propto \phi_f i_a$$

- Since the **field current** i_f and the **field flux** ϕ_f are constant, the torque τ is directly proportional to the armature current i_a ,

$$\tau(t) = K_T i_a(t)$$

where K_T is the **motor's torque constant**, the unit is $N \cdot m/A$.

- When the armature is rotating, a voltage proportional to the product of the magnetic flux ϕ_f and angular velocity ω is induced in the armature.

$$v_b \propto \phi_f \omega$$

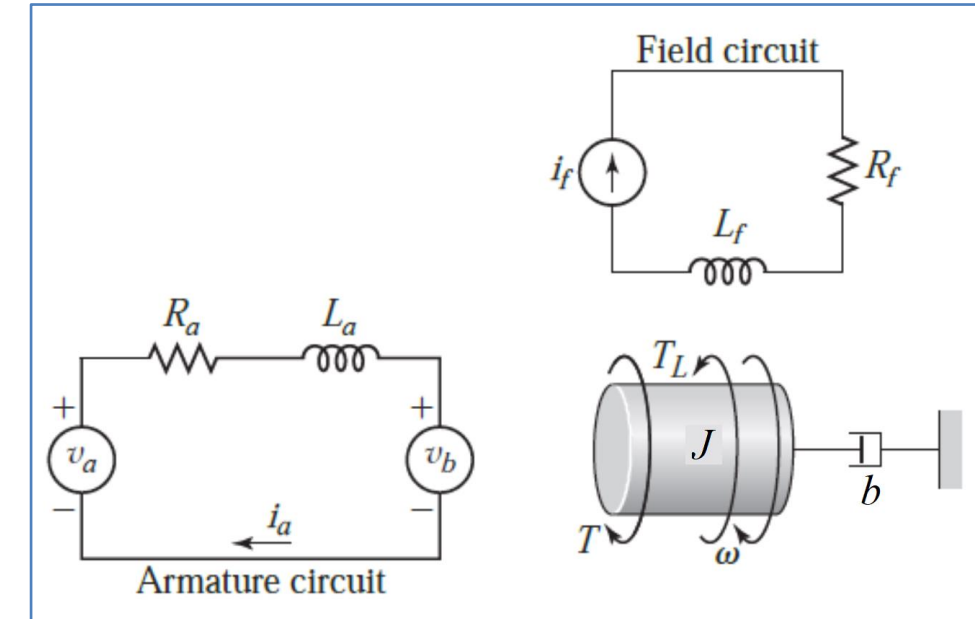
- Since the **field current** i_f and the **field flux** ϕ_f are constant, the induced voltage v_b is directly proportional to the angular velocity ω ,

$$v_b(t) = K_b \omega(t)$$

where v_b is the **back-emf** K_b is the **back-emf constant**. the unit is $V \cdot s/rad$.

- The values of K_b and K_T are **identical** if they represented in the following **SI units**:

$$K_b \left[\frac{V \cdot s}{rad} \right] = K_T \left[\frac{N \cdot m}{A} \right]$$



Modeling of Armature-Controlled DC Motor

□ Differential Equation Model

- The **speed** of an armature-controlled DC motor is controlled by the **armature voltage** v_a .
- The differential equation for the **armature circuit** is obtained by applying a KVL in the armature :

$$v_a(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt} + v_b(t)$$

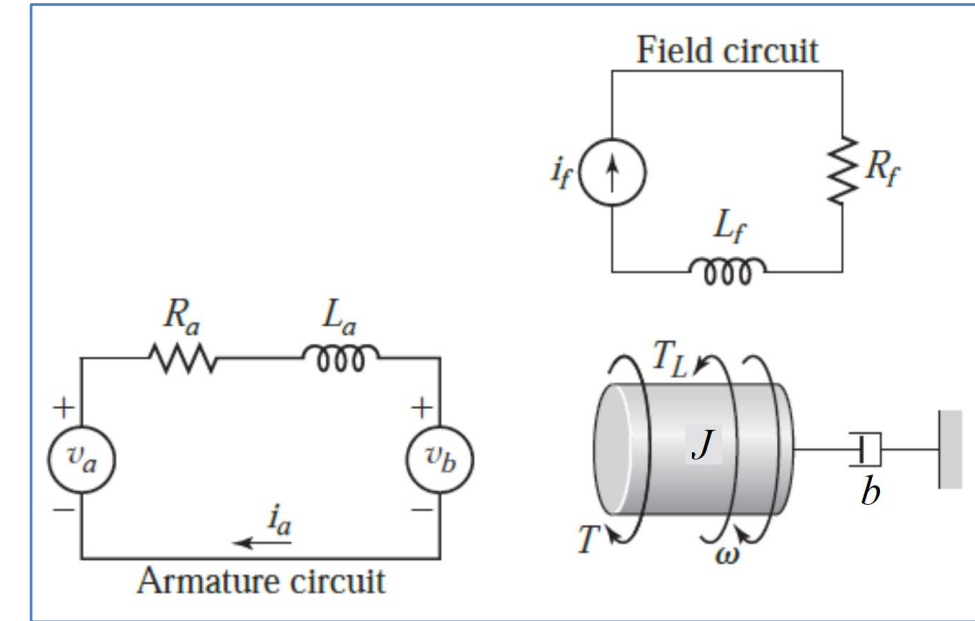
- The armature current produces the torque that is applied to the inertia and friction. Hence, from the **Newton's law** applied to the inertia J ,

$$\tau(t) - \tau_L(t) = J \frac{d\omega(t)}{dt} + b\omega(t)$$

- τ_L represents an additional torque acting on the load, other than the damping torque, which opposes the motor torque in most applications, so we have shown it acting in the direction opposite that of τ .
- These two equations along with the previously obtained two equations constitute the system model.

$$\tau(t) = K_T i_a(t)$$

$$v_b(t) = K_b \omega(t)$$



ω = Angular velocity of the motor shaft, *rad/sec*

θ = Angular displacement of the motor shaft, *rad*

τ = Torque developed by the motor, *N.m*

J = Moment of inertia of the motor and load referred to the motor shaft,

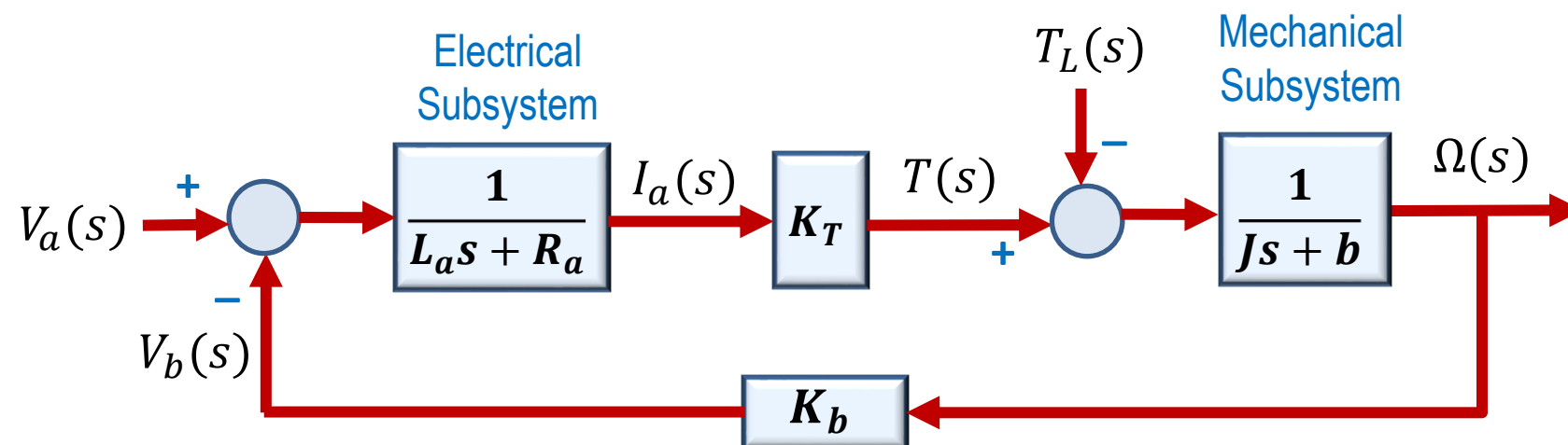
b = Viscous friction coefficient of the motor and load referred to the motor shaft,

Modeling of Armature-Controlled DC Motor

Block Diagram Model

- Block diagram model of an armature-controlled DC motor with the applied voltage $v_a(t)$ as the **input** and the motor angular velocity $\omega(t)$ as the **output** is obtained as:

$$\left\{ \begin{array}{l} v_a(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt} + v_b(t) \longrightarrow V_a(s) = (L_a s + R_a) I_a(s) + V_b(s) \longrightarrow I_a(s) = \frac{1}{L_a s + R_a} (V_a(s) - V_b(s)) \\ v_b(t) = K_b \omega(t) \longrightarrow V_b(s) = K_b \Omega(s) \\ \tau(t) - \tau_L(t) = J \frac{d\omega(t)}{dt} + b\omega(t) \longrightarrow T(s) - T_L(s) = (Js + b)\Omega(s) \longrightarrow \Omega(s) = \frac{1}{Js + b} (T(s) - T_L(s)) \\ T(t) = K_T i_a(t) \longrightarrow T(s) = K_T I_a(s) \end{array} \right.$$



Dynamics of the **electrical** and the **mechanical** subsystem are modeled as **first-order** systems.

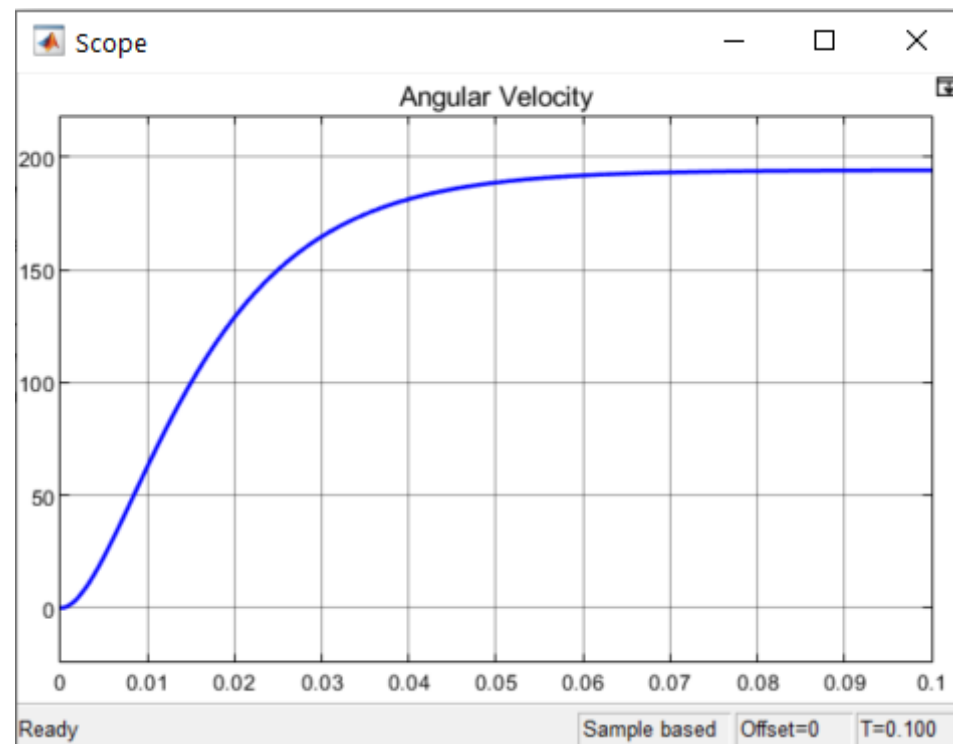
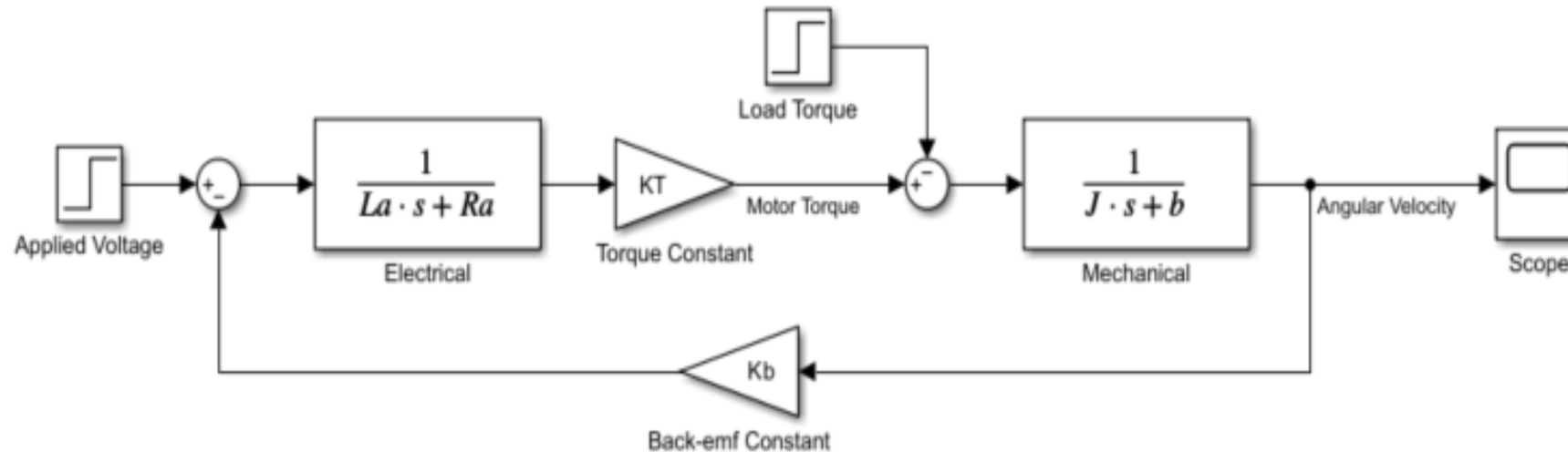
The **back-emf** acts as a **negative feedback** loop to slow down the motor's speed

Modeling of Armature-Controlled DC Motor

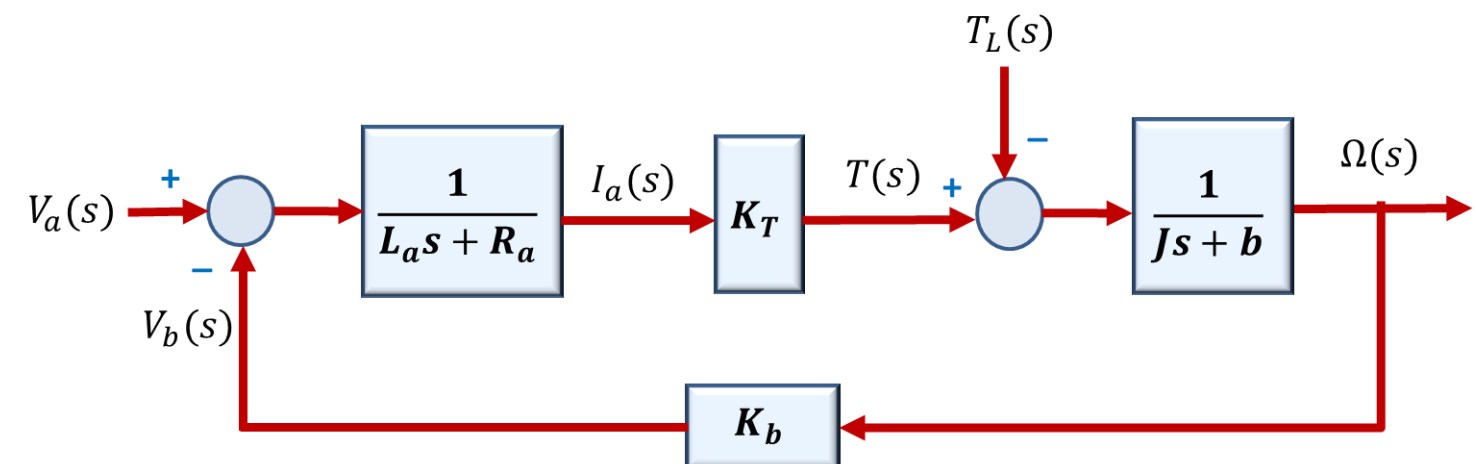
Example 1

This example shows a Simulink implementation of the DC motor block diagram model. If the applied voltage is 10V and the load torque is 0.01N.m, find the motor speed curve.

$$\begin{aligned} R_a &= 0.5\Omega, \\ L_a &= 2 \times 10^{-3}H, \\ J &= 9 \times 10^{-5}kg.m^2, \\ b &= 10^{-4}N.m.s/rad \\ K_T &= K_b = 0.05 N.m/A \end{aligned}$$



The final speed is about 196 rad/sec.



Modeling of Armature-Controlled DC Motor

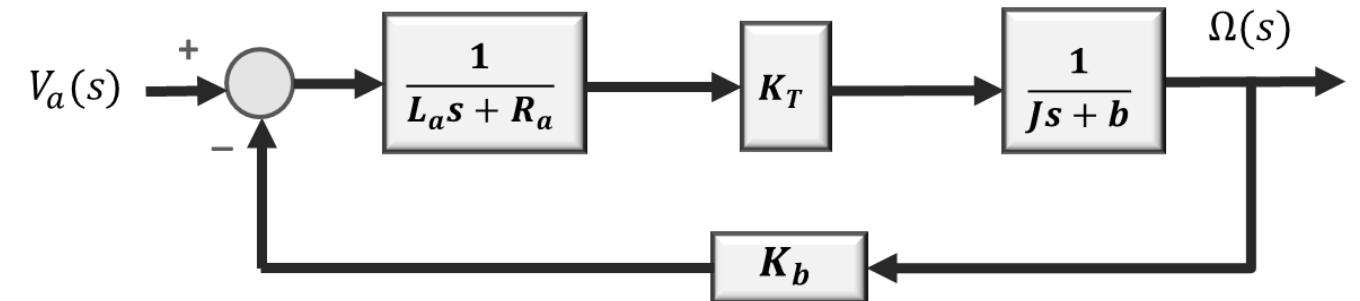
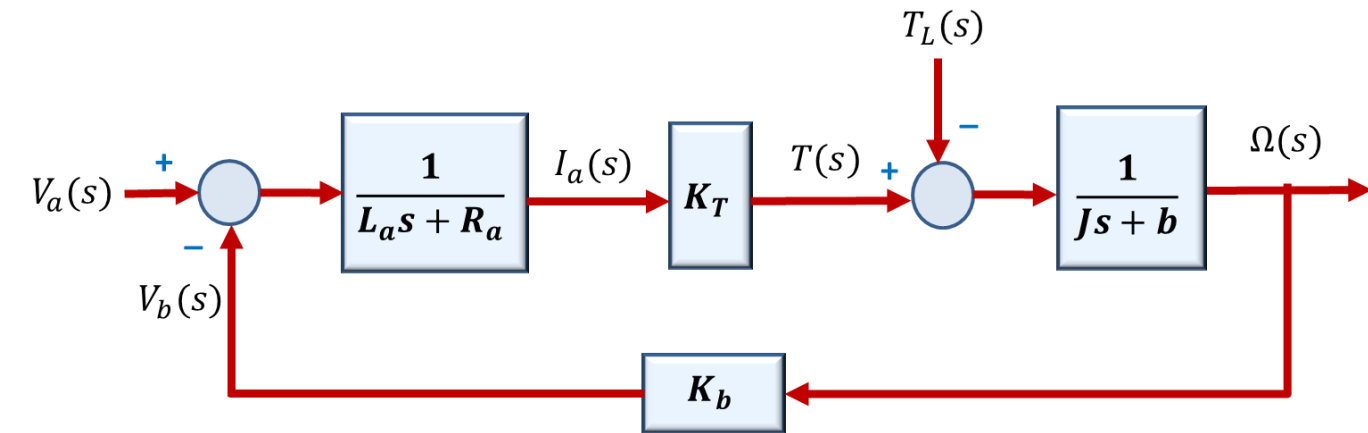
Transfer Function Model

- The **transfer function** model is obtained by finding the overall transfer function of the block diagram model.
- Since there are two independent inputs, we can apply the **superposition principle** to find the transfer function model.
- Assume a **no-load** condition and set $T_L = 0$ to obtain the **voltage-to-speed** transfer function $\Omega(s)/V_a(s)$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{G}{1 + GH} = \frac{\frac{K_T}{(L_a s + R_a)(Js + b)}}{1 + \left(\frac{K_T}{(L_a s + R_a)(Js + b)} \right) (K_b)} = \frac{K_T}{(L_a s + R_a)(Js + b) + K_T K_b}$$

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T}{L_a J s^2 + (L_a b + J R_a) s + R_a b + K_T K_b}$$

Voltage-to-Speed transfer function



Modeling of Armature-Controlled DC Motor

Transfer Function Model

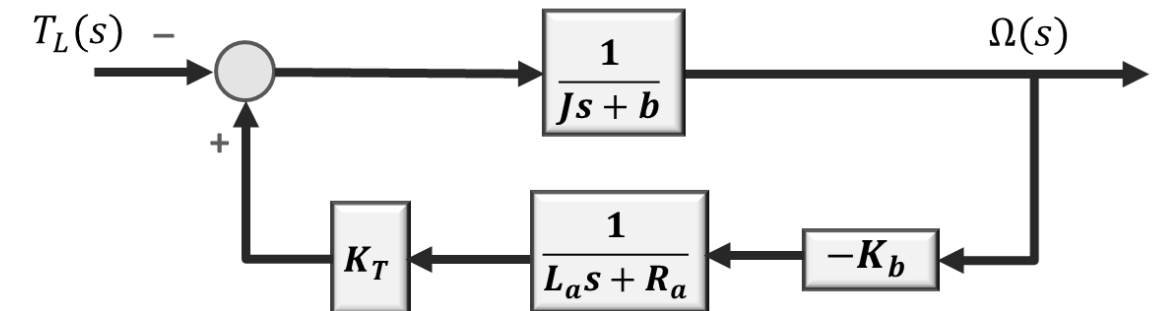
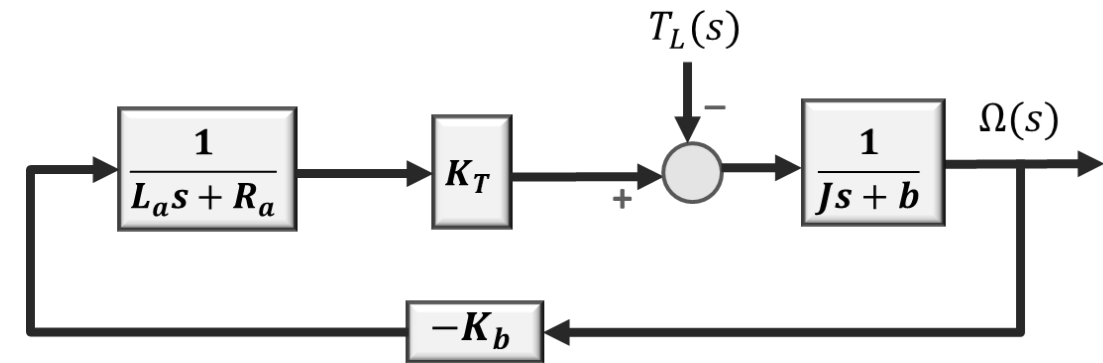
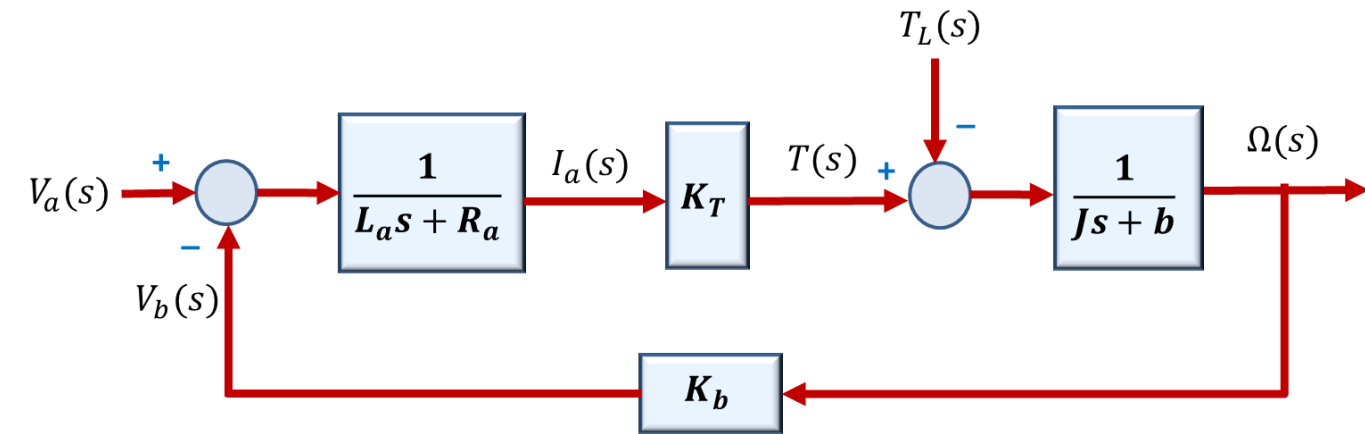
- We can derive the **load-torque-related** transfer functions to analyze the effect of the load on motor speed and current.
- Apply **superposition principle** and set $V_a = 0$ to consider the effect of **load-torque** only.
- The **Load-torque-to-speed** transfer function $\Omega(s)/T_L(s)$:

$$\frac{\Omega(s)}{T_L(s)} = -\frac{G}{1 - GH} = -\frac{\frac{1}{Js + b}}{1 - \left(\frac{1}{Js + b}\right)\left(\frac{-K_b K_T}{L_a s + R_a}\right)} = -\frac{L_a s + R_a}{(L_a s + R_a)(Js + b) + K_T K_b}$$

$$\frac{\Omega(s)}{T_L(s)} = -\frac{L_a s + R_a}{L_a J s^2 + (L_a b + J R_a) s + R_a b + K_b K_T}$$

Load-Torque-to-Speed transfer function

- The **minus sign** indicates that the speed will decrease for a positive load torque.



Modeling of Armature-Controlled DC Motor

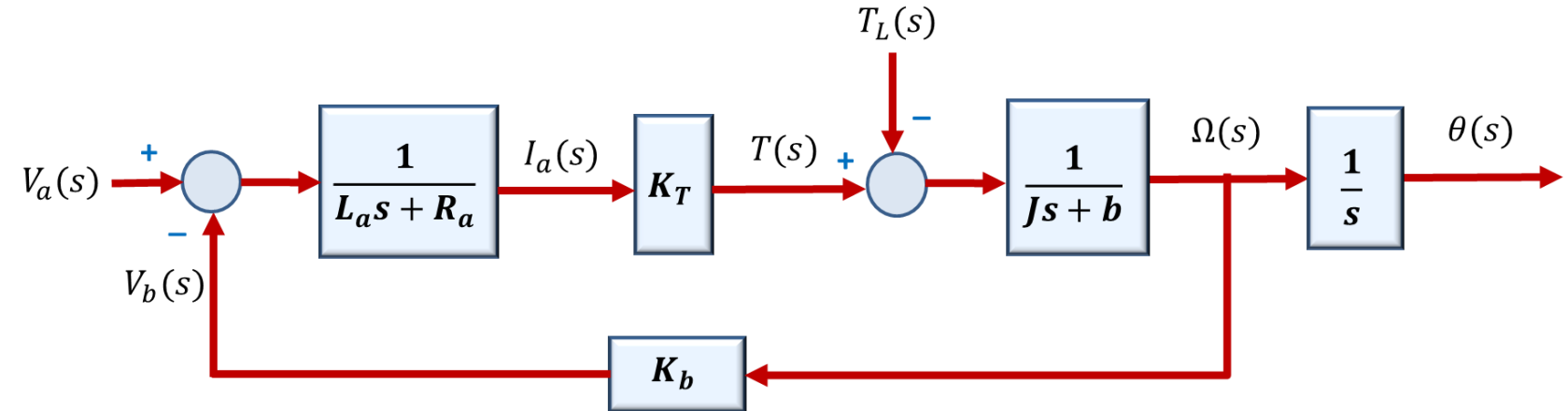
□ Transfer Function Model

- By having the **angular velocity** $\omega(t)$ of the motor shaft as an output we can easily derive the **angular displacement** $\theta(t)$ of the motor shaft and obtain the corresponding transfer function models.

$$\omega(t) = \frac{d\theta(t)}{dt} \rightarrow \Omega(s) = s\theta(s) \rightarrow \theta(s) = \frac{1}{s}\Omega(s)$$

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T}{L_aJs^2 + (L_ab + JR_a)s + R_ab + K_TK_b}$$

$$\frac{\Omega(s)}{T_L(s)} = -\frac{L_as + R_a}{L_aJs^2 + (L_ab + JR_a)s + R_ab + K_bK_T}$$



- The **voltage-to-displacement** and **load-torque-to-displacement** transfer function models are obtained as:

$$\frac{\theta(s)}{V_a(s)} = \frac{K_T}{s(L_aJs^2 + (L_ab + JR_a)s + R_ab + K_bK_T)}$$

Voltage-to-displacement transfer function

$$\frac{\theta(s)}{T_L(s)} = -\frac{L_as + R_a}{s(L_aJs^2 + (L_ab + JR_a)s + R_ab + K_bK_T)}$$

Torque-to-displacement transfer function

Modeling of Armature-Controlled DC Motor

Example 2

Given the parameters and the transfer function model of an armature-controlled DC motor. Assume the applied voltage v_a is 10V and the load torque τ_L is zero.

$$\begin{aligned}R_a &= 0.5\Omega, \\L_a &= 2 \times 10^{-3}H, \\J &= 9 \times 10^{-5}kg.m^2, \\b &= 10^{-4}N.m.s/rad \\K_T &= K_b = 0.05 N.m/A\end{aligned}$$

a) Find the time response the motor speed $\omega(t)$.

Substituting the given parameters in the transfer function model and simplify the model:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T}{L_a J s^2 + (L_a b + J R_a) s + R_a b + K_b K_T} = \frac{0.05}{18 \times 10^{-8} s^2 + 4.52 \times 10^{-5} s + 2.55 \times 10^{-3}}$$

$$\frac{\Omega(s)}{V_a(s)} = \frac{2.77 \times 10^5}{s^2 + 251s + 14160} \approx \frac{2.77 \times 10^5}{(s + 165.52)(s + 85.59)}$$

Then find the **time response** $\omega(t)$ applying **partial-fraction** expansion using MATLAB:

$$v_a(t) = 10 \quad \rightarrow \quad V_a(s) = \frac{10}{s}$$

$$\frac{\Omega(s)}{V_a(s)} = \frac{2.77 \times 10^5}{(s + 165.52)(s + 85.59)} \quad \rightarrow \quad \Omega(s) = \frac{2.77 \times 10^5}{(s + 165.52)(s + 85.59)} V_a(s) = \frac{2.77 \times 10^6}{s(s + 165.52)(s + 85.59)}$$

$$\Omega(s) = \frac{196}{s} + \frac{210}{s + 165.52} - \frac{406}{s + 85.59}$$

$$\rightarrow \quad \omega(t) = 196 + 210e^{-165.52t} - 406e^{-85.59t}$$

Modeling of Armature-Controlled DC Motor

Example 2

Given the parameters and the transfer function model of an armature-controlled DC motor. Assume the applied voltage v_a is 10V and the load torque τ_L is zero.

$$\begin{aligned}R_a &= 0.5\Omega, \\L_a &= 2 \times 10^{-3}H, \\J &= 9 \times 10^{-5}kg.m^2, \\b &= 10^{-4}N.m.s/rad \\K_T &= K_b = 0.05 N.m/A\end{aligned}$$

b) Find the final speed of the motor at the steady-state mode.

Recall the time response of motor speed:

$$\omega(t) = 196 + 210e^{-165.52t} - 406e^{-85.59t}$$

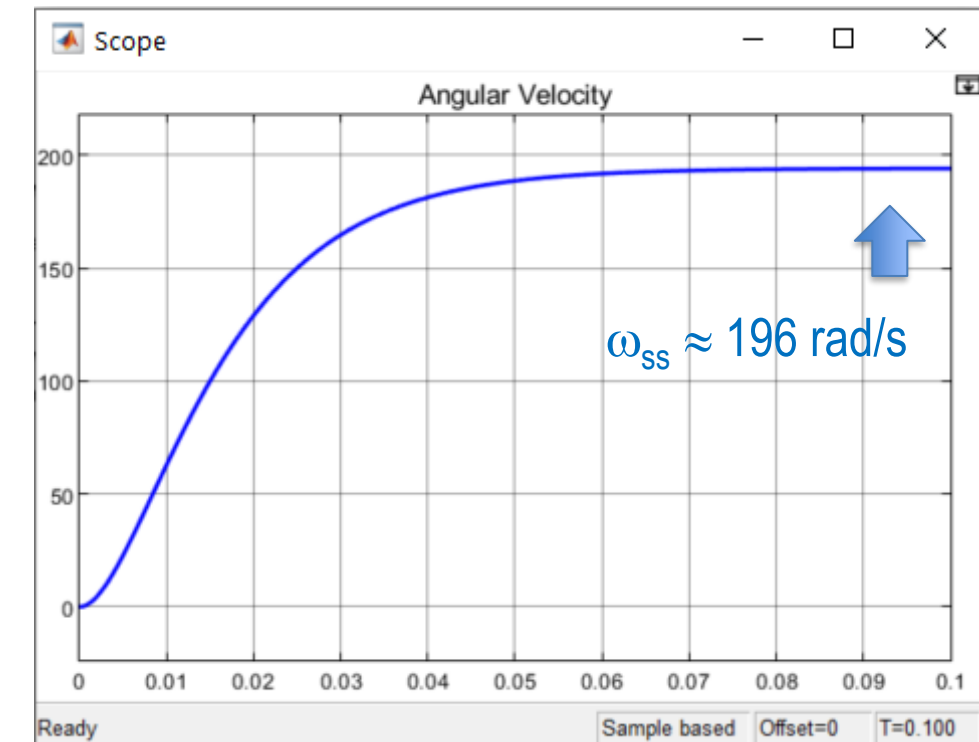
The **final speed** of the motor is:

$$\omega_{ss} = \lim_{t \rightarrow \infty} \omega(t) = \lim_{t \rightarrow \infty} (196 + 210e^{-165.52t} - 406e^{-85.59t}) = 196 \text{ rad/s}$$

We can also determine the **steady-state speed** directly from the **transfer function** model by applying **Final-Value Theorem**.

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

$$\omega_{ss} = \lim_{s \rightarrow 0} s\Omega(s) = \lim_{s \rightarrow 0} s \left(\frac{2.77 \times 10^6}{s(s + 165.52)(s + 85.59)} \right) = 195.5 \approx 196 \text{ rad/s}$$



Modeling of Armature-Controlled DC Motor

□ Simplified First-order Transfer Function Model

- The inductance L_a in the armature circuit is usually **small** and may be neglected.
- If L_a is neglected, the previously obtained second-order transfer function models reduce to a **first-order model**.
- Consider the Voltage-to-Speed transfer function:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T}{L_a J s^2 + (L_a b + J R_a) s + R_a b + K_b K_T} \xrightarrow{L_a \approx 0} \frac{\Omega(s)}{V_a(s)} = \frac{K_T}{J R_a s + R_a b + K_b K_T} \quad \text{Reduced-order Model}$$

- The reduced order model is a first-order transfer function, and it can be written in the following standard form:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T}{J R_a s + (R_a b + K_b K_T)} = \frac{\frac{K_T}{R_a b + K_b K_T}}{\frac{J R_a}{R_a b + K_b K_T} s + 1} \rightarrow \frac{\Omega(s)}{V_a(s)} = \frac{K_m}{\tau_m s + 1} \rightarrow \frac{\theta(s)}{V_a(s)} = \frac{K_m}{s(\tau_m s + 1)}$$

where,

$$K_m = \frac{K_T}{R_a b + K_b K_T} \quad \text{Motor gain}$$

$$\tau_m = \frac{J R_a}{R_a b + K_b K_T} \quad \text{Motor time constant}$$

Modeling of Armature-Controlled DC Motor

Example 3

Compares time response of the first-order model and the second-order model of the given DC motor. Applied voltage is 10V and the load torque is zero.

$$\begin{aligned}R_a &= 0.5\Omega, \\L_a &= 2 \times 10^{-3}H, \\J &= 9 \times 10^{-5}kg.m^2, \\b &= 10^{-4}N.m.s/rad \\K_T &= K_b = 0.05 N.m/A\end{aligned}$$

If L_a is neglected ($L_a = 0$), the transfer function model is reduced to a first-order model.

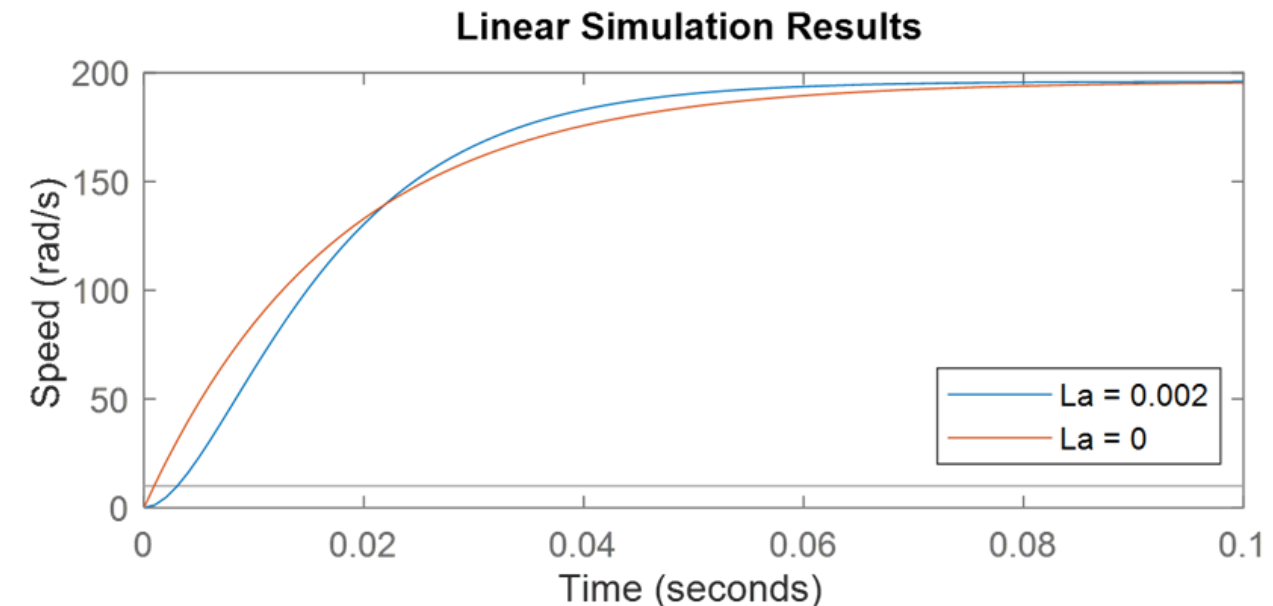
$$\frac{\Omega(s)}{V_a(s)} = \frac{K_T}{JR_as + R_ab + K_bK_T} = \frac{0.05}{4.5 \times 10^{-5}s + 25.5 \times 10^{-3}} = \frac{1.111 \times 10^3}{s + 56.67}$$

Applied voltage is 10V $\rightarrow v_a(t) = 10 \rightarrow V_a(s) = \frac{10}{s}$

$$\Omega(s) = \frac{1.111 \times 10^3}{s + 56.67} V_a(s) = \frac{1.111 \times 10^4}{s(s + 56.67)} = \frac{196}{s} - \frac{196}{s + 56.67}$$

$$\rightarrow \omega(t) = 196 - 196e^{-56.67t}$$

The results show that the simplified first-order model can provide a good approximation of the voltage-to-speed transfer function model of the DC motor.



Modeling of Armature-Controlled DC Motor

□ State Space Model

- **State space model** is obtained by considering the applied voltage $v_a(t)$ and load torque $\tau_L(t)$ as the **inputs** and the armature current $i_a(t)$ and the angular velocity $\omega(t)$ as the **outputs**.
- The state variables are defined as the armature current $i_a(t)$ and the motor speed $\omega(t)$:

$$\begin{cases} q_1(t) = i_a(t) \\ q_2(t) = \omega(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = \frac{di_a(t)}{dt} = \frac{1}{L_a} (v_a(t) - v_b(t) - R_a i_a(t)) \\ \dot{q}_2(t) = \frac{d\omega(t)}{dt} = \frac{1}{J} (\tau(t) - \tau_L(t) - b\omega(t)) \end{cases}$$

$$\begin{aligned} v_a(t) &= R_a i_a(t) + L_a \frac{di_a(t)}{dt} + v_b(t) \\ \tau(t) - \tau_L(t) &= J \frac{d\omega(t)}{dt} + b\omega(t) \\ v_b(t) &= K_b \omega(t) \\ \tau(t) &= K_T i_a(t) \end{aligned}$$

State-variable equations are obtained as:

$$\begin{cases} \dot{q}_1(t) = \frac{1}{L_a} (v_a(t) - K_b q_2(t) - R_a q_1(t)) \\ \dot{q}_2(t) = \frac{1}{J} (K_T q_1(t) - \tau_L(t) - b q_2(t)) \end{cases}$$

The **output equations** are obtained as:

$$\begin{cases} y_1(t) = i_a(t) = q_1(t) \\ y_2(t) = \omega(t) = q_2(t) \end{cases}$$

Modeling of Armature-Controlled DC Motor

□ State Space Model

- We can represent the **state** and **output equations** in the standard **matrix-vector** form as below:

$$\begin{cases} \dot{q}_1(t) = \frac{1}{L_a} (v_a(t) - K_b q_2(t) - R_a q_1(t)) \\ \dot{q}_2(t) = \frac{1}{J} (K_T q_1(t) - \tau_L(t) - b q_2(t)) \end{cases} \quad \begin{cases} y_1(t) = i_a(t) = q_1(t) \\ y_2(t) = \omega(t) = q_2(t) \end{cases}$$

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)$$

State Equation



$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} \frac{-R_a}{L_a} & \frac{-K_b}{L_a} \\ \frac{K_T}{J} & -\frac{b}{J} \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} & 0 \\ 0 & -\frac{1}{J} \end{bmatrix} \begin{bmatrix} v_a(t) \\ \tau_L(t) \end{bmatrix}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t)$$

Output Equation



$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_a(t) \\ \tau_L(t) \end{bmatrix}$$

- The state-space representation is more appropriate to model **MIMO** systems.

Modeling of Armature-Controlled DC Motor

Example 4

Consider the DC servomotor system. Assume that the armature inductance is negligible $L_a \approx 0$. Obtain the transfer function $\theta_2(s)/E_a(s)$

Modeling the electrical subsystem:

From a KVL in the armature circuit we have:

$$e_a(t) = R_a i_a(t) + e_b(t) \rightarrow E_a(s) = R_a I_a(s) + E_b(s) \quad (\text{Eqn. 1})$$

Modeling the mechanical subsystem:

Find equivalent moment of inertia and damping coefficient in the motor shaft:

$$J_{eq} = J_r + \left(\frac{N_1}{N_2}\right)^2 J_L \quad b_{eq} = \left(\frac{N_1}{N_2}\right)^2 b_L$$

The equation of motion in the motor shaft:

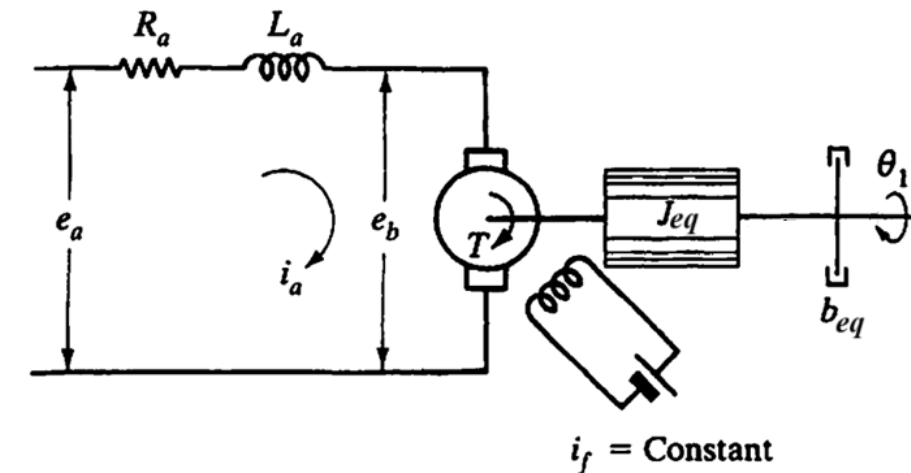
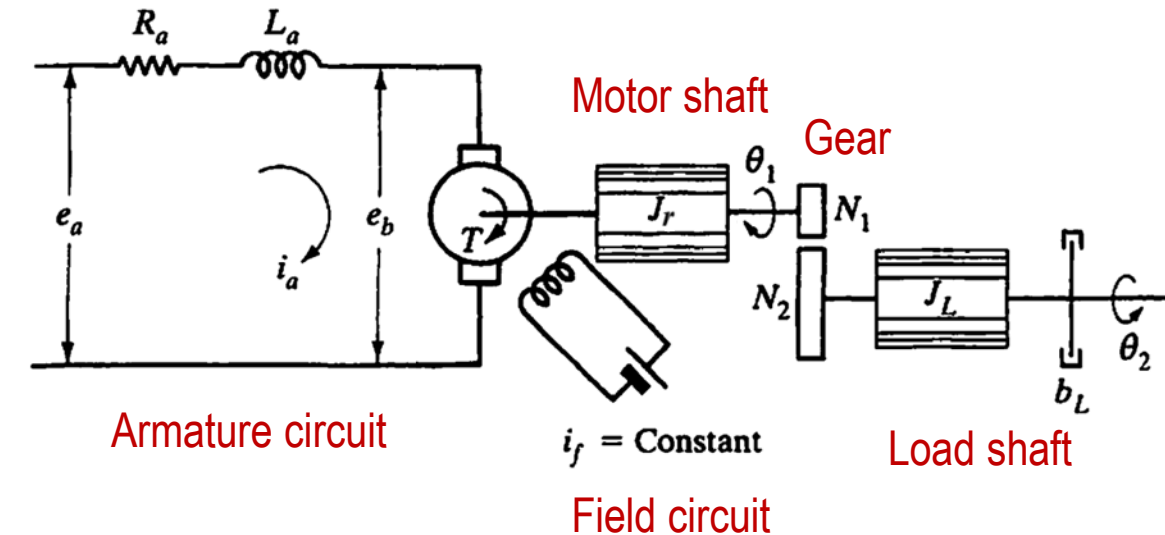
$$\tau(t) - b_{eq} \dot{\theta}_1(t) = J_{eq} \ddot{\theta}_1(t) \rightarrow T(s) = J_{eq} s^2 \theta_1(s) + b_{eq} s \theta_1(s)$$

$$T(s) = (J_{eq} s^2 + b_{eq} s) \theta_1(s) \quad (\text{Eqn. 2})$$

The equations of the developed torque $\tau(t)$ and the induced voltage $e_b(t)$

$$\tau(t) = K_T i_a(t) \rightarrow T(s) = K_T I_a(s) \quad (\text{Eqn. 3})$$

$$e_b(t) = K_b \omega_1(t) \rightarrow E_b(s) = K_b \Omega_1(s) = K_b s \theta_1(s) \quad (\text{Eqn. 4})$$



Modeling of Armature-Controlled DC Motor

Example 4

Consider the DC servomotor system. Assume that the armature inductance is negligible $L_a \approx 0$. Obtain the transfer function $\theta_2(s)/E_a(s)$

Having the equations to find the transfer function:

$$E_a(s) = R_a I_a(s) + E_b(s) \quad (\text{Eqn. 1})$$

$$T(s) = (J_{eq}s^2 + b_{eq}s)\theta_1(s) \quad (\text{Eqn. 2})$$

$$T(s) = K_T I_a(s) \quad (\text{Eqn. 3})$$

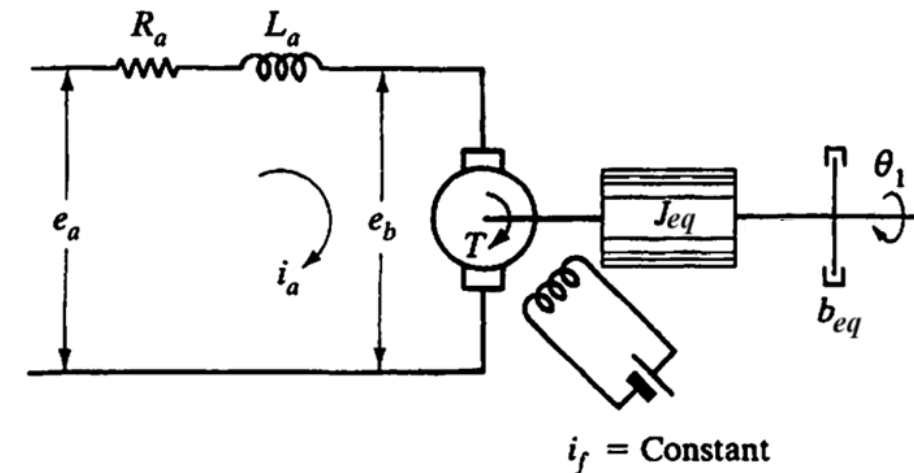
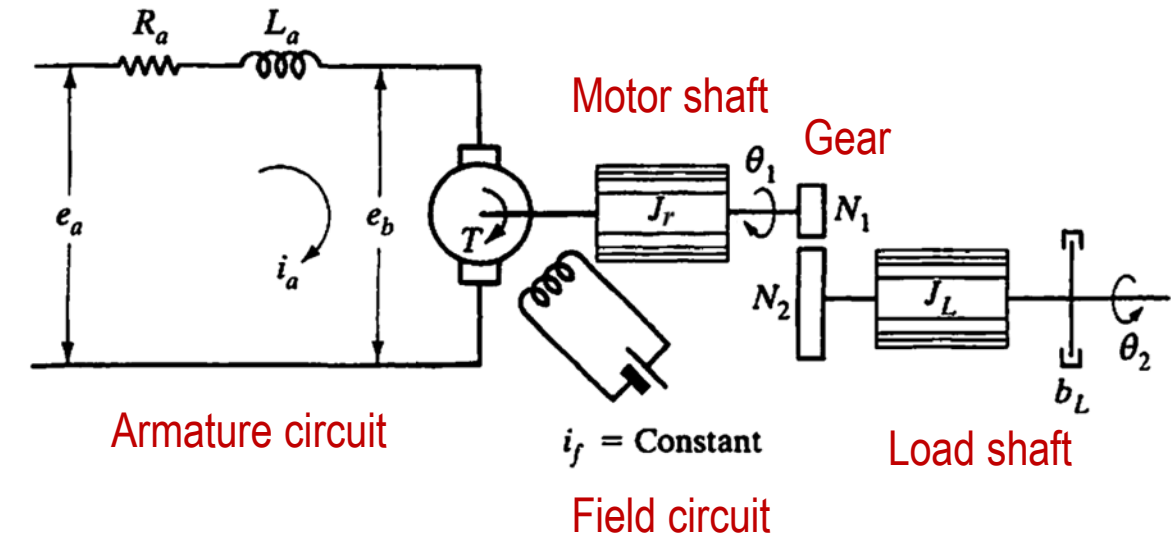
$$E_b(s) = K_b s \theta_1(s) \quad (\text{Eqn. 4})$$

Combine Eqn. (2) and (3) to find $I_a(s)$:

$$I_a(s) = \frac{T(s)}{K_T} = \frac{(J_{eq}s^2 + b_{eq}s)\theta_1(s)}{K_T}$$

Substitute $I_a(s)$ and $E_b(s)$ in Eqn. (1):

$$E_a(s) = \frac{R_a(J_{eq}s^2 + b_{eq}s)\theta_1(s)}{K_T} + K_b s \theta_1(s) = \left(\frac{R_a J_{eq}s^2 + (R_a b_{eq} + K_T K_b)s}{K_T} \right) \theta_1(s)$$



Modeling of Armature-Controlled DC Motor

Example 4

Consider the DC servomotor system. Assume that the armature inductance is negligible $L_a \approx 0$.
Obtain the transfer function $\theta_2(s)/E_a(s)$

$$E_a(s) = \left(\frac{R_a J_{eq} s^2 + (R_a b_{eq} + K_T K_b) s}{K_T} \right) \theta_1(s)$$

Since $\theta_1 = \frac{N_2}{N_1} \theta_2$

$$E_a(s) = \left(\frac{R_a J_{eq} s^2 + (R_a b_{eq} + K_T K_b) s}{K_T} \right) \frac{N_2}{N_1} \theta_2(s)$$

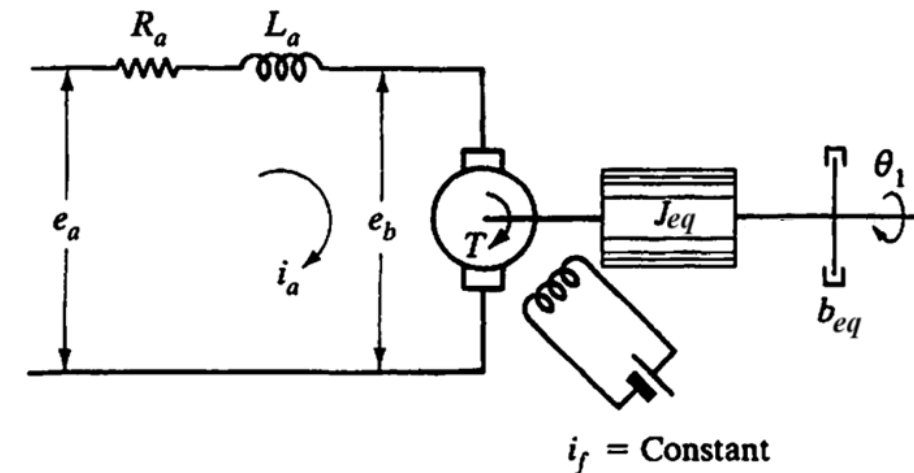
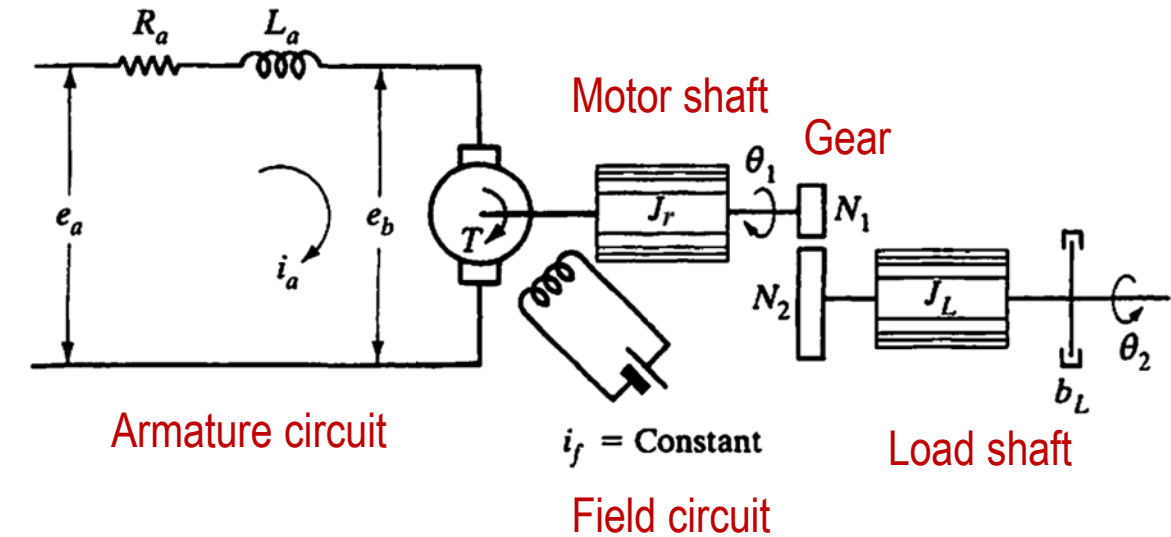
Form the transfer function $\theta_2(s)/E_a(s)$:

$$\frac{\theta_2(s)}{E_a(s)} = \frac{\frac{N_1}{N_2} K_T}{R_a J_{eq} s^2 + (R_a b_{eq} + K_T K_b) s}$$

where,

$$J_{eq} = J_r + \left(\frac{N_1}{N_2} \right)^2 J_L$$

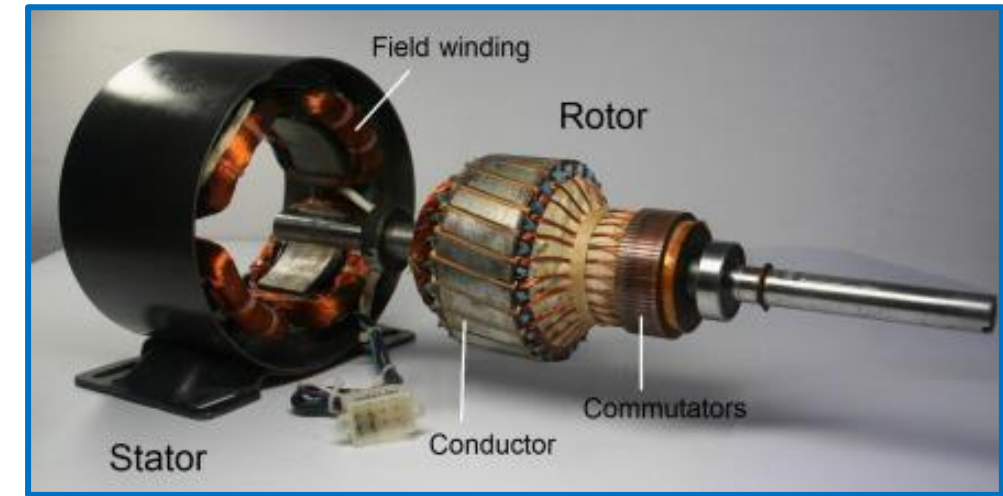
$$b_{eq} = \left(\frac{N_1}{N_2} \right)^2 b_L$$



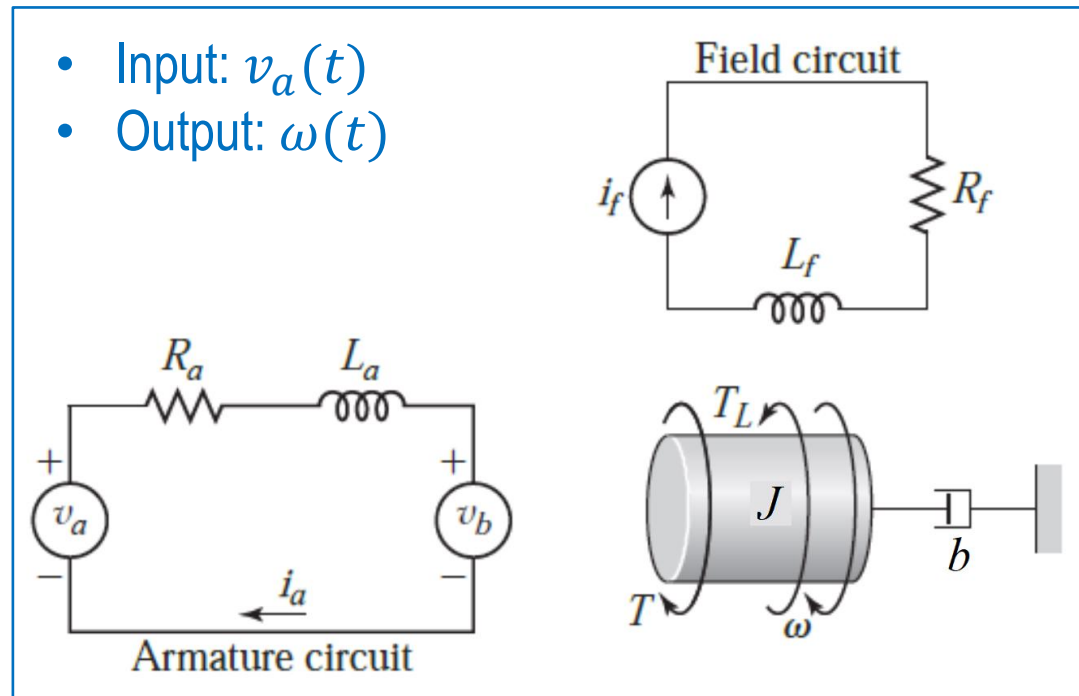
Modeling of Electromechanical Systems

□ Control of DC Motor

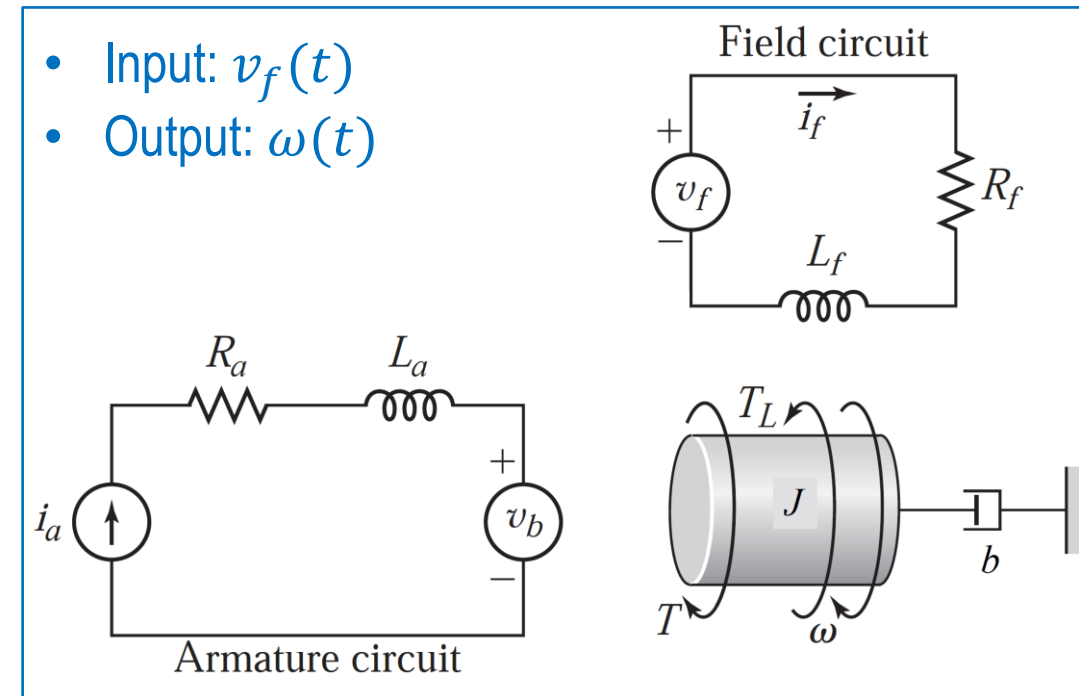
- Two general methods to control speed of DC motor:
 - Armature-Controlled DC Motor**
 - Field current i_f and field flux ϕ_f is constant.
 - Using permanent magnet
 - Field-Controlled DC Motor**
 - Armature current i_a is constant



Armature-Controlled DC Motor



Field-Controlled DC Motor

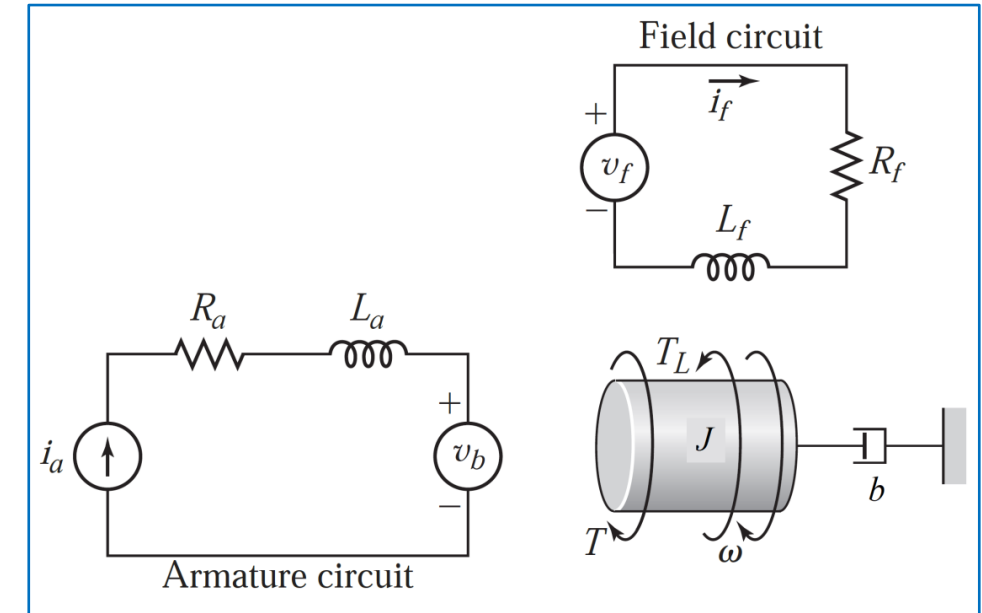


Modeling of Field-Controlled DC Motor

□ Differential Equation Model

- The **armature current** i_a is constant.
- The **voltage** v_f is applied to field circuit, whose **inductance** and **resistance** are L_f and R_f .
- The differential equation for the **field circuit** is obtained by applying a KVL:

$$v_f(t) = R_f i_f(t) + L_f \frac{di_f(t)}{dt}$$



- The torque is applied to the inertia and friction. Hence, from the **Newton's law** applied to the inertia J ,

$$\tau(t) - \tau_L(t) = J \frac{d\omega(t)}{dt} + b\omega(t)$$

- Since the **field circuit** is stationary, there is **no back-emf voltage** in the **field circuit**.
- The **torque** τ developed by the motor is a nonlinear function of the **field current** i_f . Often a linear approximation of torque is used around the operation point. Here, K_T is the **slope of the torque curve** at the operation point.

$$\tau(t) = K_T i_f(t)$$

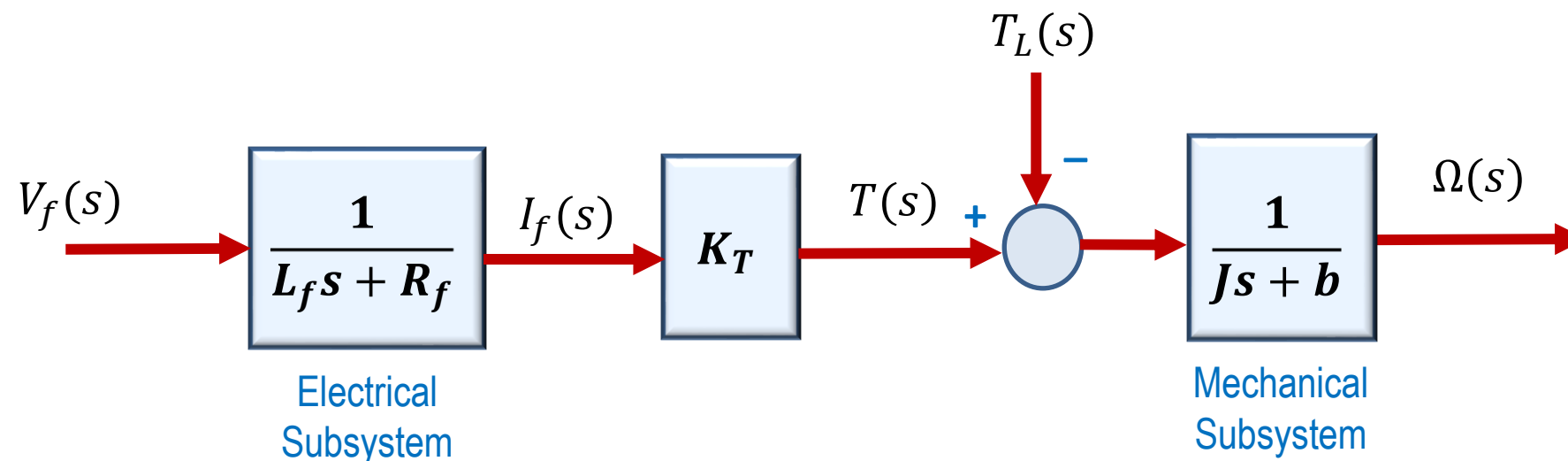
- These two differential equations along with the torque equation constitute the system model.

Modeling of Field-Controlled DC Motor

Block Diagram Model

- Block diagram model of a **field-controlled** DC motor with the **field voltage** $v_f(t)$ as the **input** and the motor **angular velocity** $\omega(t)$ as the **output** is obtained as:

$$\left\{ \begin{array}{l} v_f(t) = R_f i_f(t) + L_f \frac{di_f(t)}{dt} \longrightarrow V_f(s) = (L_f s + R_f) I_f(s) \longrightarrow \boxed{I_f(s) = \frac{1}{L_f s + R_f} V_f(s)} \\ \tau(t) - \tau_L(t) = J \frac{d\omega(t)}{dt} + b\omega(t) \longrightarrow T(s) - T_L(s) = (Js + b)\Omega(s) \longrightarrow \boxed{\Omega(s) = \frac{1}{Js + b} (T(s) - T_L(s))} \\ \tau(t) = K_T i_f(t) \longrightarrow \boxed{T(s) = K_T I_f(s)} \end{array} \right.$$



Dynamics of the **electrical** and the **mechanical** subsystem are modeled as **first-order** systems.

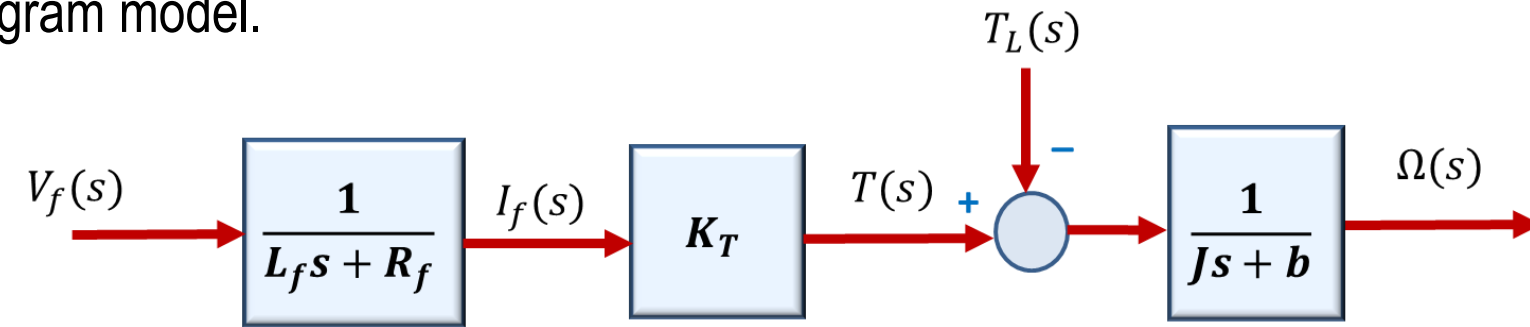
The model has **no feedback loop** because field circuit is stationary and has **no back-emf**.

Modeling of Field-Controlled DC Motor

Transfer Function Model

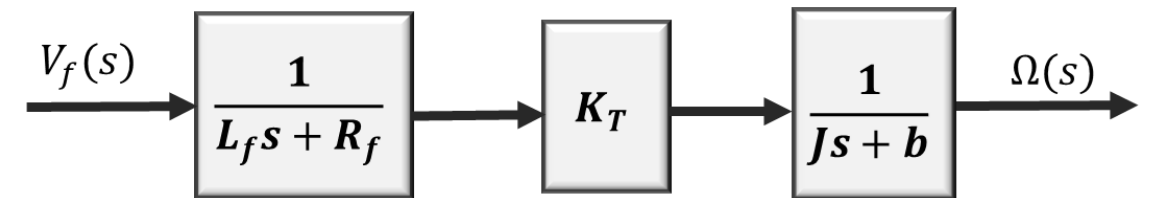
- Transfer function models can be determined from block diagram model.

- Since there are two independent inputs, we can apply the superposition principle to find the transfer function model.



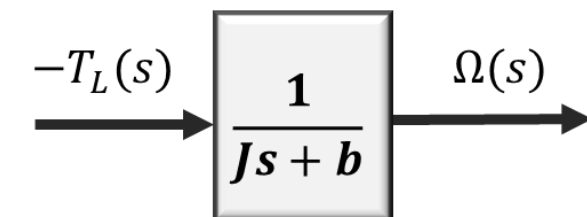
- Assume a **no-load** condition and set $T_L = 0$ to obtain the voltage-to-speed transfer function $\Omega(s)/V_f(s)$:

$$\frac{\Omega(s)}{V_f(s)} = \frac{K_T}{(L_f s + R_f)(J s + b)}$$



- The Load-torque-to-speed transfer function $\Omega(s)/T_L(s)$ by setting the $V_f = 0$:

$$\frac{\Omega(s)}{T_L(s)} = \frac{-1}{J s + b}$$



Modeling of Field-Controlled DC Motor

□ State Space Model

- **State space model** is obtained by considering the applied voltage $v_f(t)$ and load torque $\tau_L(t)$ as the **inputs** and the field current $i_f(t)$ and the angular velocity $\omega(t)$ as the **outputs**.
- The state variables are defined as the field current $i_f(t)$ and the motor speed $\omega(t)$:

$$\begin{cases} q_1(t) = i_f(t) \\ q_2(t) = \omega(t) \end{cases} \Rightarrow \begin{cases} \dot{q}_1(t) = \frac{di_f(t)}{dt} = \frac{1}{L_f} (v_f(t) - R_f i_f(t)) \\ \dot{q}_2(t) = \frac{d\omega(t)}{dt} = \frac{1}{J} (\tau(t) - \tau_L(t) - b\omega(t)) \end{cases}$$

$$\begin{aligned} v_f(t) &= R_f i_f(t) + L_f \frac{di_f(t)}{dt} \\ \tau(t) - \tau_L(t) &= J \frac{d\omega(t)}{dt} + b\omega(t) \\ \tau(t) &= K_T i_f(t) \end{aligned}$$

State-variable equations are obtained as:

$$\begin{cases} \dot{q}_1(t) = \frac{1}{L_f} (v_f(t) - R_f q_1(t)) \\ \dot{q}_2(t) = \frac{1}{J} (K_T q_1(t) - \tau_L(t) - b q_2(t)) \end{cases}$$

The **output equation** is obtained as:

$$\begin{cases} y_1(t) = i_f(t) = q_1(t) \\ y_2(t) = \omega(t) = q_2(t) \end{cases}$$

Modeling of Field-Controlled DC Motor

□ State Space Model

- We can represent the **state** and **output equations** in the standard **matrix-vector** form as below:

$$\begin{cases} \dot{q}_1(t) = \frac{1}{L_f} (v_a(t) - R_f q_1(t)) \\ \dot{q}_2(t) = \frac{1}{J} (K_T q_1(t) - \tau_L(t) - b q_2(t)) \end{cases} \quad \begin{cases} y_1(t) = i_f(t) = q_1(t) \\ y_2(t) = \omega(t) = q_2(t) \end{cases}$$

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)$$

State Equation



$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} \frac{-R_f}{L_f} & 0 \\ \frac{K_T}{J} & -\frac{b}{J} \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L_f} & 0 \\ 0 & -\frac{1}{J} \end{bmatrix} \begin{bmatrix} v_f(t) \\ \tau_L(t) \end{bmatrix}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t)$$

Output Equation



$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_f(t) \\ \tau_L(t) \end{bmatrix}$$

- The state-space representation is equivalent to the three transfer function models, which is more appropriate to model **MIMO** systems.

Modeling of Field-Controlled DC Motor

Example 5

A certain rotational system has an inertia $J = 50\text{kg.m}^2$ and a viscous damping constant $b = 10\text{Ns/m}$.

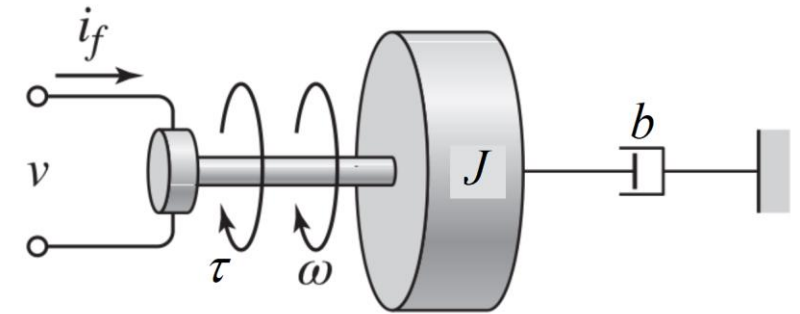
The torque $\tau(t)$ is applied by an electric motor.

The equation of motion the mechanical subsystem is

$$50\omega'(t) + 10\omega(t) = \tau(t)$$

The voltage $v(t)$ is applied to the motor. The model of the motor's field current i_f in amperes is

$$0.001i_f'(t) + 5i_f(t) = v(t)$$



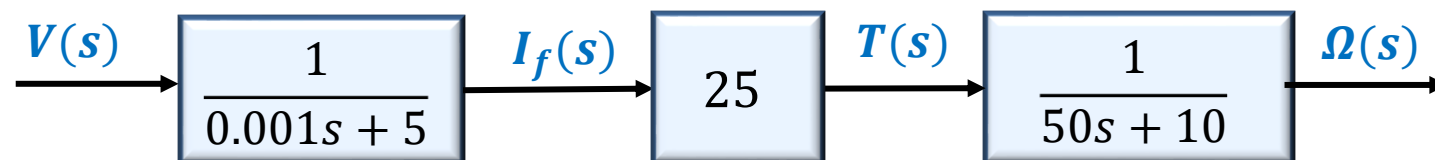
The torque-current relationship is $\tau(t) = 25i_f(t)$. Suppose the applied voltage is $v(t) = 10\text{V}$.

a) Determine the transfer function of mechanical and electrical subsystems and show the system in a block diagram model.

Transfer function of the mechanical and electrical subsystems are

$$50\omega'(t) + 10\omega(t) = \tau(t) \xrightarrow{\text{Laplace Transform}} 50s\Omega(s) + 10\Omega(s) = T(s) \longrightarrow \boxed{\frac{\Omega(s)}{T(s)} = \frac{1}{50s + 10}} \quad \text{Mechanical Subsystem}$$

$$0.001i_f'(t) + 5i_f(t) = v(t) \xrightarrow{\text{Laplace Transform}} 0.001sI_f(s) + 5I_f(s) = V(s) \longrightarrow \boxed{\frac{I_f(s)}{V(s)} = \frac{1}{0.001s + 5}} \quad \text{Electrical Subsystem}$$



Modeling of Field-Controlled DC Motor

Example 5

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The torque $\tau(t)$ is applied by an electric motor.

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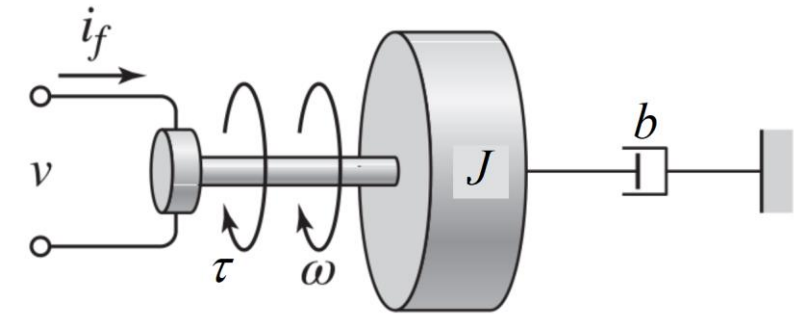
$$50\omega'(t) + 10\omega(t) = \tau(t)$$

The voltage $v(t)$ is applied to the motor. The model of the motor's field current i_f in amperes is

$$0.001i_f'(t) + 5i_f(t) = v(t)$$

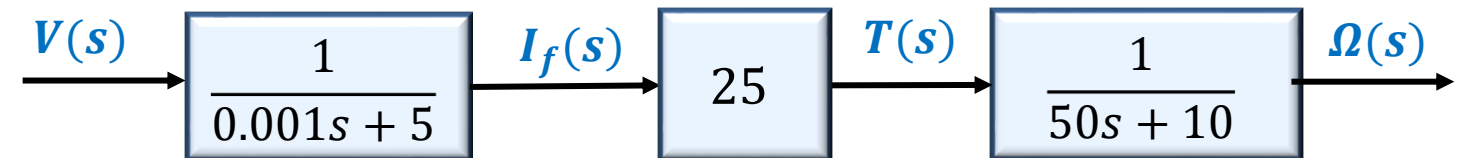
The torque-current relationship is $\tau(t) = 25i_f(t)$. Suppose the applied voltage is $v(t) = 10V$.

b) Determine the steady-state speed of the inertia.



First find the overall transfer function:

$$\frac{\Omega(s)}{V(s)} = \frac{25}{(0.001s + 5)(50s + 10)}$$



The steady-state speed is obtained using the **final-value theorem**:

$$\lim_{t \rightarrow \infty} \omega(t) = \lim_{s \rightarrow 0} s\Omega(s) \rightarrow \omega(\infty) = \lim_{s \rightarrow 0} s \left(\frac{25}{(0.001s + 5)(50s + 10)} \right) \left(\frac{10}{s} \right) = 5 \text{ rad/s}$$

Modeling of Field-Controlled DC Motor

Example 5

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The torque $\tau(t)$ is applied by an electric motor.

The equation of motion the mechanical subsystem is

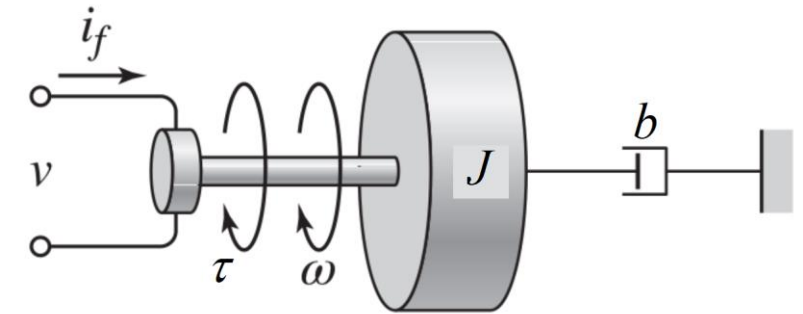
$$50\omega'(t) + 10\omega(t) = \tau(t)$$

The voltage $v(t)$ is applied to the motor. The model of the motor's field current i_f in amperes is

$$0.001i_f'(t) + 5i_f(t) = v(t)$$

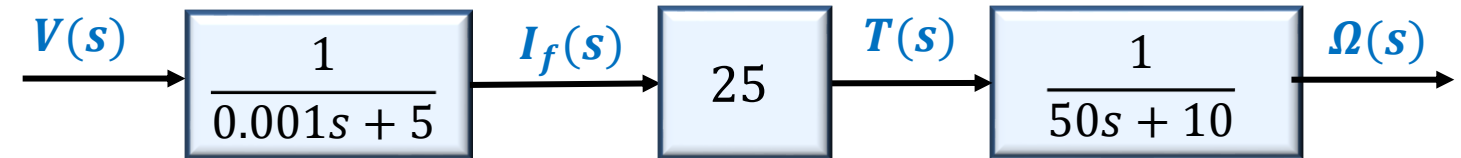
The torque-current relationship is $\tau(t) = 25i_f(t)$. Suppose the applied voltage is $v(t) = 10V$.

c) Determine the steady-state value of the current and the torque.



The steady-state value of the **current** is obtained using the **final-value theorem**:

$$\lim_{t \rightarrow \infty} i_f(t) = \lim_{s \rightarrow 0} s I_f(s) \rightarrow i_f(\infty) = \lim_{s \rightarrow 0} s \left(\frac{1}{0.001s + 5} \right) \left(\frac{10}{s} \right) = 2 \text{ A}$$



The steady-state value of the **torque**:

$$\tau(t) = 25i_f(t) \rightarrow \tau(\infty) = 25(2A) = 50 \text{ N.m}$$

THANK YOU